

Chapter 134 — Continent Urinary Diversion

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HISTORY OF CONTINENT URINARY DIVERSION

- **Objective of bladder substitution:** storage of urine plus volitional voiding — either spontaneously per urethra or by intermittent catheterisation through a robust, stable channel with a reliable continence mechanism.

Ureterosigmoidostomy and the ileal conduit

- **Ureterosigmoidostomy is the oldest continent diversion;** first bowel diversion was **Simon, 1852** — ureters brought into the rectum of an exstrophy patient by needle-and-suture fistula; patient died of sepsis at 12 months.
- **Antireflux tunnelled ureterosigmoid anastomosis** (Leadbetter, early 1950s) markedly reduced obstruction and ascending pyelonephritis, but long-term **hyperchloraemic acidosis, ureteric obstruction, pyelonephritis and secondary malignancy at the implantation site** persisted.
- **Bricker popularised the ileal conduit in 1950**, building on **Zaayer (1911)**; it separated urine from stool, is technically simple and reliable, and remains **the commonest urinary diversion worldwide** after cystectomy.
- **Conduit late complications emerged in the 1970s:** skin and appliance problems, peristomal hernia, stomal stenosis, pyelonephritis, stones, ureteric obstruction and renal deterioration.
 - **Attributed to high-pressure reflux of infected urine** — yet **non-refluxing colon conduits had the same complications**, undermining reflux as the sole cause.

Continent cutaneous diversion

- **Gilchrist, 1950:** first human continent cutaneous diversion — **ileocaecal valve as continence mechanism, distal ileum as catheterisable stoma**; little attention at the time.
- **Kock, 1982:** reintroduced the concept from the continent ileostomy — **intussuscepted nipple valve** for continence and antireflux.
 - Kock established **complete detubularisation and double-folding to the most spherical shape** — the cornerstone of all modern cutaneous and orthotopic reservoirs.
 - **Skinner** adopted it for adult cystectomy from 1982, reporting his initial large series by 1987.
- **Central problem of cutaneous diversion** = a reliable, durable, easily catheterisable **effluent continence mechanism**; every design shares **stones, difficult catheterisation, peristomal hernia and stomal leakage**, often needing revision.

Orthotopic diversion

- **Tizzoni and Poggi, 1888:** first experimental orthotopic diversion — isolated ileal loop between ureters and urethra in a dog, which was continent and had three pregnancies.
- **Lemoine, 1913:** first human orthotopic reconstruction — converted a ureterosigmoidostomy by transposing sigmoid to the urethral stump.
- **Camey and Le Duc, 1979:** pioneering clinical orthotopic substitution in men; the original Camey used **intact (tubular) ileum → high-pressure reservoir and unacceptable continence**.
- **Subsequent designs:** caecal and ileocaecal neobladders (1986), sigmoid reservoir (Reddy, 1987), **Hautmann W-neobladder (1988)**, **Studer pouch (1989)**, right colon (Månsson and Colleen, 1990), Camey II (1990), hemi-Kock (1991), extraserous-lined ureteric tunnel (Abol-Enein and Ghoneim, 1993), stomach

neobladder (1998), T pouch (1998), Padua pouch — many with long follow-up showing good renal preservation and continence.

- **Orthotopic has overtaken continent cutaneous:** natural voiding, simpler construction, less revision surgery. **Cutaneous remains favoured for pre-existing incontinence or urethral disease.**
- **Women initially excluded** — continence was believed to need an intact bladder neck, and urethral recurrence was feared. **Mid-1990s:** anatomical dissection and clinical experience showed **continence with a low-pressure reservoir and the urethra alone**, and oncological safety.
- **Orthotopic neobladder most closely resembles the native bladder:** no appliance, volitional urethral voiding, no intermittent catheterisation for most, and no efferent-limb problems.

Adoption and uptake

- **Rapid growth in the 1990s–2000s:** from 1986, orthotopic substitution rose to >75% of cystectomies at some centres.
- **Wide variation:** neobladder rates across eight large international academic centres ranged 6–51%.
- **Population-level uptake is low and falling:** US national data — only ~9% received continent diversion (more in teaching hospitals and patients <60), falling yearly after 2008; a high-volume centre halved its neobladder rate 2000–2005; National Surgical Quality Improvement Program (NSQIP) and National Cancer Database data confirm the decline; Sweden ~14%.
- **Why uptake is low:**
 - **Patient factors:** older, more obese, more comorbid, socioeconomic challenges; quality-of-life studies **not consistently favouring neobladder.**
 - **Surgeon factors:** **little residency exposure; no economic incentive** (United States) for a longer operation; surgeons can easily **frame options to steer patients** towards the diversion the surgeon prefers.
 - **Robotics:** push to fully intracorporeal diversion further reduces continent diversion.

LANDMARK TRIALS — CONTINENT DIVERSION RATES IN ROBOTIC VERSUS OPEN CYSTECTOMY TRIALS

Trial	Country	Randomisation	Continent diversion	Point
RAZOR	United States	Open vs robotic cystectomy with extracorporeal diversion	21%	Low despite all surgeons fellowship-trained or bladder-cancer focused
iROC	United Kingdom	Open vs robotic cystectomy with totally intracorporeal diversion	11%	Intracorporeal approach halves the rate further

- **Medical eligibility far exceeds use:** **80–90% of men and ~75% of women** undergoing cystectomy are medically candidates for a neobladder.
 - Ulm series: only a small minority had medical contraindications, and very few refused.
 - With **standardised preoperative counselling**, very few eligible patients chose a conduit for personal reasons.
 - **PROMETRICS study:** orthotopic neobladder 28.8%, continent cutaneous 2.6%; neobladder associated with **younger age, male sex, fewer comorbidities.**
- **Guidelines:** American Urological Association (AUA)/Society of Urologic Oncology (SUO) and European Association of Urology (EAU) — **counsel every cystectomy patient on all three diversion options; refer** an eligible, interested patient to a surgeon experienced in orthotopic diversion if the treating surgeon is not.

KEY POINTS: HISTORIC EVOLUTION OF CONTINENT URINARY DIVERSION

- Three paths: **conduit → continent cutaneous → orthotopic.**

- Neobladder: volitional voiding, no appliance, **self-catheterisation in only a minority**.
- **80–90% of men and 75% of women** undergoing cystectomy are potential orthotopic candidates.
- Continent cutaneous diversion = continent option **when orthotopic is contraindicated**; favoured with pre-existing incontinence or urethral disease.

Concept — Every advance in this history solves the **same two problems: reservoir pressure and outlet reliability**. Detubularisation fixed the reservoir, the native rhabdosphincter fixed the outlet, and the gap between high eligibility and low uptake is now a problem of surgeon training, incentive and counselling rather than of patient suitability.

BASIC PRINCIPLES OF CONTINENT URINARY DIVERSION

- **Principle 1 — a reliable outlet:**
 - **Orthotopic:** adequate **external sphincter** and a **non-obstructed urethra**.
 - **Continent cutaneous:** a robust, stable **efferent catheterisable channel** with a **reliable continence mechanism**.
- **Principle 2 — low pressure (compliance) throughout filling:** achieved by **complete detubularisation** and **folding into a sphere**.
 - **Sphere = greatest volume for a given surface area**.
 - Kock double-folding and S- and W-shaped reservoirs **abolish coordinated high-pressure peristaltic contractions**.
- **Principle 3 — adequate volume: ≥500 mL once mature**, for reasonable voiding intervals.
 - **All bowel stretches over time** given adequate outflow resistance; **ileum stretches more than colon**.
 - Standard **44 cm of ileum** double-folded (Kock; also Studer and T pouch): **<200 mL initially → 500–600 mL at low pressure within the first year**.
 - **A large initial volume is not needed** to reach an adequate mature voiding volume.

FIGURE 134.1

- **Computed tomography (CT) of the pelvis, Studer ileal neobladder at 1 year:** the pouch has become **spherical** with maturation.

KEY POINTS: THREE BASIC PRINCIPLES OF CONTINENT DIVERSION CONSTRUCTION

- Orthotopic: **healthy urethra + adequate external sphincter**.
- Cutaneous: **adequate efferent channel + reliable continence mechanism**.
- Reservoir **completely detubularised and spherical**.
- Mature storage volume **≥500 mL at low pressure**.

Concept — Continence in any continent diversion is a **pressure balance: low reservoir pressure against adequate outlet resistance**. Detubularisation lowers the numerator, and the sphincter or efferent valve supplies the denominator; failure of either side causes leakage.

PATIENT SELECTION

- **Nearly every cystectomy patient is at least a potential continent diversion candidate;** start by excluding contraindications.

CONTRAINDICATIONS TO CONTINENT DIVERSION

Category	Orthotopic diversion only	All continent diversions
Absolute	• Invasive cancer at the urethral margin • Significant urethral stricture disease	• Severe renal dysfunction • Severe hepatic dysfunction • Physical or mental inability to self-catheterise
Relative / counselling factors	Poor external sphincter; palpable bladder-neck or anterior vaginal wall tumour; prostatic stromal invasion; hostile irradiated sphincter	Frailty, severe comorbidity, poor cognition, difficult social circumstances, prior radiation, prior bowel resection

- **Warn every patient that intraoperative findings may force an alternative diversion** — rare, but up to ~10%.
- **Stoma siting before surgery:**
 - Authors prefer the **umbilicus for continent cutaneous stomas** — easy to conceal, locate and manage.
 - **Also mark every patient for an ileal conduit as the final back-up.**
 - **Conduit site rules:** free of fat creases standing, sitting and lying; away from umbilicus and scars; **as lateral as possible while still passing through the middle of the rectus** — failing this raises **parastomal hernia** risk.

FIGURE 134.2 ALGORITHM FOR PATIENT SELECTION

Each step must be passed; any failure → **do NOT offer orthotopic diversion.**

eGFR >40 mL/min → **intact, functional external sphincter** → **adequate available bowel** → **no** very frail state or severe liver, cardiac or psychological disease → sex-specific oncological check:

- **Men:** no cancer at the **distal prostatic urethra** → **offer orthotopic diversion.**
- **Women:** no palpable tumour at the **anterior vagina** or **cervical invasion** → **willing and able to learn self-catheterisation** → **offer orthotopic diversion.**
- **Trap:** the algorithm places the self-catheterisation gate only on the female limb, but the text requires **all orthotopic candidates of both sexes** to be willing and able to self-catheterise.

Oncologic Factors**Risk for Urethral Recurrence in Male Patients**

- **Primary oncological contraindication** = urothelial carcinoma at the urethral margin on intraoperative frozen section.
- **Prostatic urethral involvement is the key risk factor for urethral recurrence.** Pooled data: anterior urethral recurrence after cystectomy ~8%.
- **USC long-term series** (men, urethra retained): overall urethral recurrence ~7% at 5 years, ~9% at 10 years; median onset 2 years (range to 13 years).
 - **Risk rises with depth of prostatic involvement:** none (lowest) < carcinoma in situ (CIS) or mucosal/ductal involvement < stromal invasion (highest).
- **Prostatic stromal invasion usually travels with nodal disease** — most such men are node-positive, so **diagnose it preoperatively and give neoadjuvant chemotherapy.** Even in this group urethral recurrence is relatively uncommon, and lower in those with a neobladder.

MALE URETHRAL RECURRENCE — THE CONTESTED PREDICTORS

Factor	USC series	Mayo series	Verdict
Prostatic urethral / stromal involvement	Significant	Significant	Consistent — the dominant risk factor
Tumour multifocality	Not significant	Significant	Discordant
Bladder CIS	Not significant	Not significant	Not a predictor (other series variable)

- **The neobladder itself appears protective against urethral recurrence:**
 - USC: lower 5-year urethral recurrence with neobladder than cutaneous diversion, both **without** prostatic involvement and **with stromal invasion**.
 - Mayo: same protective effect; **persists after adjusting for pathology**.
 - **Meta-analysis (>6000 patients): hazard ratio (HR) 0.44** favouring orthotopic diversion.
 - Mechanism unknown — speculated **continued urine flow and bowel-altered urine**.

Preoperative prostatic biopsy versus intraoperative frozen section

- **Technique:** at transurethral resection (TUR), **deep TUR biopsies of the prostatic urethra at 5 and 7 o'clock, lateral to the verumontanum**; mandatory if prostatic mucosa looks suspicious. Some advocate this routinely before orthotopic diversion.

EVIDENCE: PREOPERATIVE TUR PROSTATIC BIOPSY VS INTRAOPERATIVE FROZEN SECTION

Study finding	Favours
Prospective comparison: frozen section predicted urethral recurrence better than any preoperative parameter , including TUR biopsy	Frozen section
TUR biopsy missed a large proportion of prostatic involvement on final pathology; a third of TUR-positive stromal cases had no residual disease at cystectomy	Frozen section
Orthotopic series: TUR–frozen section correlation poor; negative predictive value for final margin — TUR 99.4% vs frozen section 100%	Frozen section (marginally)
Routine TUR biopsy: a negative biopsy correlated >99% with negative frozen section — may obviate frozen section	TUR biopsy

- **Authors' practice: no routine repeat TUR** solely to exclude prostatic involvement (others likewise, with comparable outcomes).
- **Counselling:** prostatic mucosal, ductal or stromal involvement → explain the higher recurrence risk of a retained urethra and weigh it against the benefit of a neobladder.
 - **Stromal invasion → neoadjuvant chemotherapy** (high nodal risk).
 - Not a chemotherapy candidate, or **persistent prostatic urethral disease at surgery → cutaneous diversion + concomitant or delayed urethrectomy**.
- **Mandatory urethral surveillance:** periodic urethral wash cytology, urethroscopy, and attention to **urethral discharge or spotting**.

Risk for Urethral Recurrence in Females

- **Historically routine urethrectomy** in female cystectomy; neobladder success in men prompted urethral preservation.
- **Female continence does not need the bladder neck:** excellent continence with **urethral division just distal to the bladder neck**.
- **Preserve the urethra in women whatever the diversion** — distal urethrectomy **devascularises the clitoris**, impairing arousal and desire.

- **Two pivotal pathological studies** justified urethra-sparing cystectomy:
 - **Pathological review of female cystectomy specimens:** a quarter had bladder-neck tumour; **about half of those had urethral tumour; every patient with a negative bladder neck had a negative urethra — no skip lesions.** Risk rose with grade, stage and nodes; **CIS did not predict.** Vaginal wall invasion was rare but always accompanied by bladder-neck disease and often urethral extension.
 - **Long-term cohort of women with all stages:** urethral involvement ~2%, **always with bladder-neck involvement;** no link with CIS or multifocality.
 - **Conclusion: preserve the urethra if neither preoperative bladder-neck biopsy nor intraoperative proximal urethral frozen section shows tumour or severe atypia.** Other studies confirm **bladder-neck tumour as the most significant risk factor.**
- **Prospective validation:** intraoperative proximal urethral frozen section **correctly identified every case** of urethral involvement on final pathology.
- **Clinical outcomes:** first USC female neobladders — no urethral recurrences at medium-term follow-up; another prospective series — only isolated urethral recurrences (one primary squamous cell carcinoma, one CIS of the trigone), **no isolated vaginal wall recurrences;** uterine or cervical invasion was rare and **always suspected on preoperative imaging.**

Omitting urethral frozen section

- **Rationale:** margin-recurrence data come largely from historical conduit cohorts; a **positive urethral frozen section is the commonest reason to abort a planned orthotopic diversion,** so omission may **reduce disparities in receipt of continent diversion.**
- **Evidence:** a series without frozen section found **no association of positive urethral margin with recurrence-free or overall survival** — provocative, short follow-up; CIS versus invasive margins may differ; needs validation.
- **AUA guideline: verify a negative urethral margin before constructing an orthotopic diversion.** Authors send a **urethral margin from the final specimen after extirpation.**

URETHRAL RECURRENCE: MEN VERSUS WOMEN

Feature	Men	Women
Dominant risk factor	Prostatic urethral involvement, worst with stromal invasion	Bladder-neck tumour; anterior vaginal wall involvement
Best preoperative assessment	TUR biopsy optional (5 and 7 o'clock); not mandatory	Bimanual examination under anaesthesia; bladder-neck biopsy
Definitive intraoperative test	Frozen section of the urethral margin — both sexes	
CIS as predictor	Bladder CIS not predictive	CIS not predictive
Skip lesions	—	None: negative bladder neck → negative urethra
Disqualifying finding	Cancer at distal prostatic urethra / positive margin	Palpable tumour at bladder neck or anterior vagina; cervical invasion

Locally Advanced Tumor Stage

- **Two reasons for reluctance:** (1) local recurrence might damage the neobladder; (2) these patients were assumed to die of distant disease and not benefit.
- **Clinical staging is unreliable after neoadjuvant chemotherapy** because of treatment-related pelvic change.

- **USC series (>1000 patients, >10-year follow-up): local pelvic recurrence ~7% overall — ~6% organ-confined node-negative vs ~13% extravesical or node-positive; about half with extravesical disease and a third node-positive were disease-free at 5 years.**
 - Contemporary update: recurrence-free survival (RFS) unchanged over time, **5-year overall survival improved (~51% → 62%).**
 - So local recurrence is uncommon even when advanced, and many are **long-term survivors who can benefit from a neobladder.**
- **When local recurrence does occur, it rarely disables the neobladder:** in the Ulm series most recurrences affected the **upper tracts** (commonest), then neobladder, then bowel; only one neobladder removed (intestinal fistula). Elsewhere, conversion to conduit was rarely needed.
- **Adjuvant radiotherapy after neobladder (45–50.4 Gy):** well tolerated, no high-grade gastrointestinal toxicity.
- **Authors do not deny orthotopic diversion for extravesical extension or nodal disease found at surgery. Exception: grossly positive margins at the pubic bone, rectum or pelvic side wall (rare).**

KEY POINTS: ONCOLOGIC FACTORS IN PATIENT SELECTION

- **CIS, multifocality or extravesical disease do not preclude** orthotopic diversion if the urethral frozen section is negative.
- Men: strongest risk factor for urethral recurrence = **prostatic stromal invasion** on final pathology.
- Women: strongest risk factor = **bladder-neck or anterior vaginal wall cancer**, best assessed by **bimanual examination under anaesthesia.**
- **Intraoperative frozen section accurately identifies urethral involvement;** preoperative prostatic or bladder-neck biopsy is **not mandatory.**
- **Local pelvic recurrence <10%** with meticulous lymphadenectomy; rarely affects neobladder function.

Concept — Oncological selection turns on one anatomical question: is there tumour at the retained outlet?

In men the route is the prostatic urethra, in women the bladder neck and anterior vagina, and the frozen section settles it for both. Stage, nodes and CIS elsewhere change prognosis, not the choice of diversion.

Patient-Related Factors

- **Factors to weigh:** general health, social circumstances, baseline renal function, healthy urethra and functioning sphincter, manual dexterity, prior pelvic radiation, prostate surgery or bowel resection — and **patient preference** about incontinence, self-catheterisation and an appliance.
- **Bias runs both ways:** patients gravitate to the "natural" neobladder, surgeons to the **simplest, quickest, most familiar** diversion. Every diversion has problems, of different kinds; **honest counselling**, ideally with contact with other patients.
- **Ileal conduit is best for the frail elderly, poor overall health, high surgical risk, difficult social circumstances or poor cognition.**

Age

- **Chronological age alone is not a contraindication.** The elderly take **longer to regain continence** and have more **mild stress incontinence**, but **ultimately reach day and night continence similar to the young.**
- **Deciders in the elderly:** comorbidity (renal, cardiac, pulmonary), cognition, dexterity, social support. A conduit is **easier for a carer**; an active, healthy, independent older patient remains a reasonable neobladder candidate.

Renal Function

- **Compromised renal function is one of the most important contraindications.**
- **Mechanism:** bowel mucosa **reabsorbs urea, potassium and chloride** and **excretes sodium and bicarbonate** → **acid load for the kidneys** → with poor function, **hyperchloraemic metabolic acidosis**, dehydration, uraemia, nausea and bone loss.
- **Evidence that moderate impairment is safe:**
 - Short-term eGFR (estimated glomerular filtration rate) fall was smaller after continent diversion than conduit, and **no different between baseline eGFR ≥ 60 and 40–59 mL/min.**
 - Matched neobladder cohort (baseline 40–59 vs ≥ 60 mL/min): GFR stayed at or above baseline in a similar proportion; the lower-eGFR group took **longer** to lose 10 mL/min.
 - **No difference in renal decline with continent cutaneous diversion** either; **no evidence that orthotopic diversion causes progressive renal dysfunction** vs conduit when baseline function is normal.
- **Threshold:** serum creatinine **<1.7–2.2 mg/dL (150–200 $\mu\text{mol/L}$)** or **eGFR >35–40 mL/min/1.73 m².**
- **Obstructive uraemia is often reversible:** tumour-related upper-tract obstruction raises creatinine transiently and should improve after cystectomy — factor it into counselling.
- **All patients:** hydration, avoid nephrotoxins, maintain acid–base balance with **oral sodium bicarbonate.**

Body Habitus

- **Obesity is not a contraindication**, but urethral suturing and a thick mesentery are harder.
- **Commonest technical reason to convert from planned orthotopic = short or fatty mesentery** (more common with high body mass index (BMI)) → pouch will not reach the urethra.
- **Mobilisation manoeuvres:** deeper mesenteric division at the **distal** resection site; **discard a bowel segment at the proximal site**; **reduce table flexion / Trendelenburg**; gentle gradual traction of the reservoir to the urethra.
- **Paradox:** the obese may be **better served by orthotopic diversion**, because a functional stoma through a thick abdominal wall is difficult.

Manual Dexterity and Willingness to Do Self-Catheterization

- **Self-catheterisation needed** (occasional or routine) in **10–20% of men and ~30% of women with a neobladder**, and in **all continent cutaneous patients.**
- **Failure to catheterise** → **perforation, stones, renal decline, infection.** Assess self-care capacity; an **enterostomal therapist** consultation helps.
- **Poor candidates for any continent diversion: multiple sclerosis, quadriplegia, very frail or mentally impaired** — they will eventually need carers, and an appliance is far less burdensome for carers and the health system than a catheterisable pouch.
- **Retention is unpredictable and may appear years later** → **every orthotopic candidate must be willing and able to self-catheterise**; especially women (higher need, technically harder). Nearly all can learn; **few decline a neobladder once educated.**

Urethral Stricture Disease or External Sphincter Damage

- **Severe urethral stricture = contraindication to orthotopic diversion** (both sexes).
- **Poor sphincter, motivated man** → concomitant or delayed **artificial urinary sphincter.**
- **Significant stress incontinence, woman** → concomitant **Burch or pubourethral sling**, at the cost of **likely dependence on intermittent catheterisation.**

Prior Pelvic Radiation

- **Context:** salvage cystectomy after failed bladder radiotherapy or radiation for another pelvic cancer. Some surgeons use a **transverse colon conduit** (outside the radiation field).
- **Irradiated patients have more complications even with a conduit**, particularly those needing intervention; yet ileal conduit ureteroileal complications were low enough that **transverse colon is not required**.
- **Neobladder after full-dose radiation (≥ 60 Gy):** operative outcomes and complications similar to non-irradiated patients; early complications tracked with **age and American Society of Anesthesiologists (ASA) score, not diversion type**; large series showed no difference in 30- and 90-day complications or RFS; daytime continence similar. **Caveat: highly selected**, least sphincter damage.
- **Conditions for a neobladder after radiation: no preoperative incontinence or stricture, no fistula, no severe bowel radiation damage.** Selected women also suitable.
- **Worst for the rhabdosphincter: high-dose prostate radiation (external beam or brachytherapy) and vaginal implants for cervical cancer** scar more than external-beam radiation for bladder or other cancers; **brachytherapy seeds migrate into levators and urogenital diaphragm**.
- **Assessment:** preoperative cystoscopy of sphincter mucosa, but **final decision is intraoperative** (urethra, ureters, bowel). **Always agree the second-choice diversion before surgery.**

Prior Prostate Surgery or Bowel Resection

- **After radical prostatectomy:** difficult dissection at the old vesicourethral anastomosis but usually feasible; assess the urethra endoscopically before and during surgery.
 - **Predictor = continence before cystectomy:** most men continent before regained good continence (0–1 pad); those with poor continence often needed an artificial sphincter. No rectal injuries or anastomotic strictures.
 - **Flexible cystoscopy during apical dissection** locates the prior anastomosis so that all bladder is removed.
- **Multiple prior bowel resections:** removing a further **45–60 cm of small bowel** risks **chronic diarrhoea or short bowel syndrome** → consider a **sigmoid neobladder**.
 - Otherwise: free all small bowel, take down adhesions, and **take down the old anastomosis and use it as one end of the reservoir** — avoids an **ischaemic segment between old and new anastomoses**.

KEY POINTS: PATIENT-RELATED FACTORS

- **Adequate renal function (eGFR ≥ 35 –40 mL/min)** recommended.
- **Older age and obesity are not contraindications.**
- **Orthotopic diversion possible after pelvic radiation or radical prostatectomy** in carefully selected men and women with a **preserved, healthy rhabdosphincter**.
- **Prior bowel resection** → **take down the old anastomosis and use that segment** to avoid devascularisation.

Concept — *Patient selection asks whether the patient can safely carry the extra physiological and practical load of a continent reservoir: the kidneys must clear the reabsorbed acid, and the patient or carer must be able to catheterise when emptying fails. Age and weight matter only through these two demands.*

CONTINENCE MECHANISM IN PATIENTS UNDERGOING CONTINENT DIVERSION

- **Continence depends on precise dissection at the prostatic apex (men) and bladder neck (women),** without compromising oncological clearance.
- **Sphincter = smooth + striated muscle; the striated rhabdosphincter is the most critical,** innervated by the pudendal nerve.

RHABDOSPHINCTER ANATOMY: WOMEN VERSUS MEN

Feature	Women	Men
Location of striated fibres	Mainly ventral and lateral to the urethra	Omega-shaped , ventral and lateral to membranous urethra and prostate — not a horizontal ring
Relation to smooth muscle / levator	Gradual transition in the middle third , intermingled fibres	Independent striated muscle, not in contact with levator ani
Primary continence zone	Mid-urethra (fluorourodynamics after neobladder)	Membranous urethra
Innervation	Pudendal branches under levator; perineal pudendal fibres enter the caudal urethra laterally beneath the urogenital diaphragm	Fine pudendal branches
Implication	Injury to the rhabdosphincter or its pudendal supply impairs continence	

SURGICAL TECHNIQUES FOR CONTINENCE PRESERVATION DURING RADICAL CYSTECTOMY

Anterior Apical Dissection in the Male Patient

- **Minimal manipulation** of periurethral tissue and its innervation; techniques mirror nerve-sparing radical prostatectomy.
- **Posterior plane:** bladder and prostate freed off the rectum; **posterior dissection must not extend distal to the prostatic apex.** If nerve sparing: take lateral pedicles (anterior internal iliac branches) → divide urethra → dissect prostate retrograde off rectum and neurovascular bundles → divide posterior pedicles last.

Apical sequence: divide anterior fibroareolar attachments → incise endopelvic fascia, sweep levators off prostate → ligate and divide superficial branch of the dorsal vein → **only slightly incise puboprostatic ligaments** → **control deep dorsal venous complex (DVC) with absorbable suture** → divide DVC close to the apex → **suspend DVC to pubic periosteum** (anterior fixation may speed continence) → incise anterior urethra just beyond the apex → **place 2-0 absorbable urethral sutures circumferentially before full division** → clamp and divide catheter → two posterior sutures through **rectourethralis and caudal Denonvilliers fascia** → divide posterior urethra → tag sutures until after the ureteroenteric anastomosis.

- **Absorbable DVC suture** avoids erosion of suture, clips or staples into the urethral anastomosis; **avoid deep bites** into the complex or levators (continence injury).
- **Sutures before full division** because the **male urethra retracts** once divided — contrast with the female urethra below.
- **Frozen section of the distal urethral (apical) margin:**
 - **Tumour at the apex** → excise stump or **total urethrectomy** → **cutaneous or conduit diversion.**
 - **Atypia or focal CIS** → send a **second, more distal margin** if feasible.
- **Nerve sparing:** benefit to continence **controversial** (retrospective, selection bias) — reasonable in a **potent man when oncologically safe.**

Preservation of the Urethra, Reproductive Organs, and Neurovascular Tissue in the Female Patient

- **Standard female cystectomy = anterior exenteration** (uterus, cervix, ovaries). In **selected lower-stage disease, preserve uterus and ovaries.**
- **Preserving uterus, anterior vagina and ligaments → eliminates vaginal fistula risk, improves sexual function, may reduce late retention.**
 - If uterus preserved: take bladder branches **distal to the uterine artery**; dissect the ureter from **beneath the uterine artery.**
- **Dissect the bladder off the anterior vaginal wall** rather than excising it wherever possible. A deep posterior or trigonal tumour may require partial anterior vaginal wall excision — **raises pouch–vaginal fistula risk but is not an absolute contraindication.**
- **Palpable tumour at the bladder neck or extending into the vaginal wall → poor neobladder candidate → en bloc urethrectomy + cutaneous diversion.**
- **Posterior pedicles:** with hysterectomy, open the posterior fornix behind the cervix; after prior hysterectomy, a **sponge stick** identifies the vaginal apex (vagina usually not entered).
- **Lateral vaginal autonomic nerves:** some credit them with continence and avoiding retention; others rely on **pudendal innervation alone** successfully. **No randomised trials**; reasonable to preserve for continence and sexual function.

Bladder-neck sequence: divide posterior pedicles → sharp dissection of bladder off anterior vagina in the correct plane (avoid entering bladder; vessel-rich) → **tug Foley balloon to mark the vesicourethral junction** → sweep fat off anterior urethra **without disturbing endopelvic fascia or periurethral tissue** → free bladder neck and first 0.5 cm of urethra → mark incision just distal to junction → remove Foley, **pedicle clamp across bladder neck** (prevents tumour spill) → divide proximal urethra → frozen section of urethral margin.

- **Male vs female urethra:** the **female proximal urethra lies above the endopelvic fascia and does not retract**, so sutures can be placed **after** removing the bladder; the **male urethra retracts**, so sutures go in **before** division.
- **En bloc anterior vaginal wall** (deep posterior tumour): incise along the anterolateral vaginal wall; **keep a rim of anterior vagina proximal to the urethra** so the vaginal and vesicourethral suture lines do not abut; close clamshell or horizontally. Complex vaginal reconstruction (skin graft, rectus myocutaneous flap) is limited by pelvic space.
- **Vaginal support:** close apex; suspend to Cooper ligament or round ligaments to prevent prolapse — authors routinely perform **sacrocolpopexy with a short strip of cadaveric mesh** to the sacral promontory, supporting the neobladder against late voiding difficulty.
- **Always interpose a vascularised omental pedicle** between vagina and neobladder, fixed to endopelvic fascia either side of the urethral stump — **separates suture lines (prevents fistula) and supports the pouch posteriorly.**

FIGURE 134.3

- **Female urethral preparation:** fat swept off endopelvic fascia; vesicourethral junction identified with the Foley balloon at the bladder neck (~20 mL of water helps); **anterior endopelvic fascia and periurethral tissue left undisturbed.**

KEY POINTS: CONTINENCE MECHANISM AND SURGICAL TECHNIQUES FOR CONTINENCE PRESERVATION

- Rhabdosphincter fibres and pudendal innervation lie **anterior and lateral to the proximal urethra** in both sexes.
- Men: **careful DVC control; avoid deep bites into pelvic floor muscles.**

- **Women:** meticulous urethral and bladder-neck dissection; **leave endopelvic fascia and levators undisturbed**; preserve reproductive organs and neurovascular tissue when feasible.

Concept — *Because the rhabdosphincter is an anterolateral, pudendally innervated structure, continence-preserving surgery comes down to staying off the anterior and lateral periurethral tissue. Every technical rule, whether controlling the DVC, avoiding deep bites or leaving the endopelvic fascia intact, protects that territory.*

TECHNIQUES FOR CONTINENT BLADDER SUBSTITUTION

Choice of Bowel Segment

- **Surgeon preference often outweighs physiology**; good orthotopic results are achievable with any segment **if the rhabdosphincter is preserved and a capacious low-pressure reservoir is built**. The ideal surgeon is fluent in several techniques.
- **Muscle distensibility** (canine model): **ileal circular (most distensible) > colonic circular > ileal longitudinal > colonic longitudinal (relatively non-distensible)**.
- **Ileum is urodynamically superior**: larger capacity, lower filling and maximum-capacity pressures, better compliance. **Stomach and sigmoid have particularly poor compliance and high pressures**.
- **Urine exposure changes ileal mucosa: mucosal atrophy** (more reliable in small than large bowel) → less mucus and less electrolyte reabsorption; epithelium also **loses its intestine-specific epigenetic profile** (de novo methylation of intestinal enhancers).
- **Authors use ileum whenever available**.

CHOICE OF BOWEL SEGMENT FOR A CONTINENT RESERVOIR

Segment	Advantages	Disadvantages	Best use
Ileum	Best compliance , lowest pressures, largest capacity; most mobile mesentery — reaches urethra ; afferent limb can reach the renal pelvis for short ureters; reliable mucosal atrophy	Vitamin B12 malabsorption (distal ileum — relevant to Indiana pouch); may be irradiated, resected or diseased (Crohn disease)	Default for orthotopic diversion
Ileum + colon	Compliance close to ileum; colon's high-pressure waves damped	Loses ileocaecal valve	Mainz-type reservoirs
Colon alone	Available when ileum is short or diseased	Less distensible; high-pressure waves → incontinence ; needs larger initial volume; less reliable atrophy	Multiple small-bowel resections; diseased ileum
Stomach	Secretes H⁺ and Cl⁻ — reverses acidosis ; acid lowers bacterial colonisation; less mucus	Poor compliance; dysuria and haematuria from acid; unfamiliar anatomy — rarely used	Renal impairment ; heavy prior radiation or bowel resection (combined with bowel)

- **Gastric + intestinal composite** offsets each segment's side effects — metabolically ideal, theoretically usable where renal function contraindicates standard continent diversion.
- **Segment isolation protects blood supply**: avoid deep mesenteric incision at the **proximal** ileal division (reservoir supply); after prior resection **use the old anastomotic site**, not a nearby new one.

Need to Prevent Reflux

- **Origin of the antireflux doctrine**: upper-tract damage from reflux in ureterosigmoidostomy and ileal conduits → belief that **high-pressure reflux of infected urine** destroys kidneys.

- **Comparative evidence is weak** (short, retrospective, small, selected); non-randomised series found **no difference in renal function or hydronephrosis** between refluxing and non-refluxing neobladders.
- **Why reflux matters less in a neobladder:** patients void by **Valsalva with sphincter relaxation** — pressure rises **equally throughout the system**, limiting back-pressure. Studer pouches freely reflux on cystogram from ~300 mL, yet long-term follow-up shows **preserved renal function and anatomy in ~95%**.
- **Every antireflux mechanism can itself obstruct.**

URETERIC IMPLANTATION: ANTIREFLUX MECHANISMS VERSUS DIRECT ANASTOMOSIS

Technique	Mechanism	Obstruction / stricture	Other problems
Kock intussuscepted nipple	Afferent intussusception, stapled	Late afferent nipple stenosis or extussusception , often >10 years, silent until bilateral hydronephrosis or renal failure	Stones on exposed staples ; technically difficult — largely abandoned
Le Duc (Camey–Le Duc) mucosal groove	Submucosal tunnel in ileum	7–29%	—
Direct ureteroileal (refluxing)	End-to-side (Leadbetter) or Wallace	<3% (3–6% in larger series)	Reflux, apparently harmless
Serous-lined extramural tunnel (Abol-Enein and Ghoneim)	Ureter in a serosal trough	~3.8% — similar to refluxing anastomoses; upper tracts stable or improved in ~96%, reflux ~3%	Needs long ureters ; cannot take dilated ureters
T pouch (T limb)	Tunnelled ileal afferent limb, flap valve	Afferent T-limb stenosis ~2% at medium term (may rise); renal function preserved in ~90%	Accepts dilated or short (high-divided) ureters

LANDMARK RANDOMISED TRIAL — REFLUXING VERSUS ANTIREFLUX NEOBLADDER

Trial	Arms	Population	Endpoints at 3 years	Verdict
USC randomised trial	Studer (refluxing direct anastomosis) vs T pouch (non-refluxing afferent limb)	484 radical cystectomy patients	Renal function and urinary tract infection: no difference	T pouch needed more diversion-related secondary procedures → antireflux adds cost without benefit

- **Longer survival demands lifelong renal monitoring**; on balance the **obstruction risk of antireflux techniques outweighs their upper-tract benefit**.

KEY POINTS: CHOICE OF BOWEL SEGMENT AND THE NEED TO PREVENT REFLUX

- **Ileum, or ileum plus colon**, has the best physiological properties.
- Isolate the segment carefully to **preserve blood supply** to pouch and bowel anastomosis.
- **Antireflux mechanism not necessary** to preserve upper tracts or prevent infection, at least in the intermediate term.

Concept — *The reflux debate is settled by where the pressure goes during voiding. A low-pressure neobladder emptied by Valsalva pressurises the whole system equally, so reflux is benign, while every valve built against it adds an obstructive failure point. **Obstruction, not reflux, is what destroys kidneys.***

General Perioperative Management

- **Same as for incontinent diversion**; outcomes improved by **Enhanced Recovery After Surgery (ERAS)** multimodal pathways.

- **ERAS Society cystectomy guideline, 2013** (largely from colorectal data): **22 items**, including **no oral mechanical bowel preparation**, minimally invasive approach, **opioid-sparing analgesia**, careful fluid management, **no postoperative nasogastric tube**, early diet and mobilisation.
- **Outcomes**: authors discharge at **3–5 days**; fewer gastrointestinal complications and shorter stay **without more readmissions**, whatever the approach; meta-analyses show **fewer complications and faster bowel recovery**.

AUA-RECOMMENDED ERAS MEASURES

#	Measure	Detail
1	Counsel	Risks of diversion-related complications and their impact on quality of life
2	Optimise performance status	Smoking cessation in all (lower morbidity, better oncological outcome); up to 70% of cystectomy patients are malnourished or frail → frailty index and cardiopulmonary exercise testing for high risk; prehabilitation (nutrition ± exercise) supported by randomised trials
3	No oral mechanical bowel preparation; no prolonged fasting — use carbohydrate loading	Faster bowel recovery, better fluid and electrolyte balance, no rise in wound infection, leak or mortality. Exception: authors still prepare the bowel when colon is used (faecal and bacterial load)
4	Strict anaesthetic protocol	Restrictive transfusion and fluids when safe, normothermia, normoglycaemia
5	μ-opioid antagonist (alvimopan)	Randomised trial: faster bowel recovery and less ileus vs placebo, no extra complications

- **Postoperative ileus is the commonest gastrointestinal complication** after diversion — prolongs stay, adds morbidity and cost. Authors' ERAS with alvimopan cut ileus from **~23% to ~7%**.

Stents, catheters and drains

- **No consensus**; most use **early ureteric stents**, with removal timing highly variable.

DRAINAGE STRATEGIES AFTER CONTINENT DIVERSION

Element	Authors' approach	Alternative approach
Ureteric stents	7-Fr single-J, tied to the urethral catheter — removed together, fewer bags	Externalised stents through the skin, drained separately
Pouch drainage	24-Fr stiff haematuria catheter per urethra (orthotopic) or as caecostomy tube (cutaneous); routine irrigation; ~3 weeks	Separate suprapubic tube
Trade-off	Simpler for patient and mobility	Less urine through the healing pouch, less nursing dependence; cumbersome, limits mobility

Follow-up sequence: pelvic drain removed when output is low → review at **~day 10** (bloods, drain removal) → **catheter out at 3 weeks** → **no routine pouchogram** unless drain output is significant → education throughout on catheter care, **pelvic floor exercises**, voiding or catheterisation technique → frequent voids at first, **intervals lengthened progressively**.

Surgical Techniques

- **Many minor modifications exist**; the following is not exhaustive.

Orthotopic Ileal Reservoirs

- **Most use 60–75 cm of terminal ileum**, detubularised and folded towards a sphere; variants differ in **folding and ureteric management**.

- **Close with continuous absorbable suture; avoid non-absorbable suture and metal staples (stones).**
Absorbable staples: awkward, inconsistent, not routine.
- **Two most popular worldwide: Hautmann W and Studer** — both simple and allow **direct ureteroileal anastomosis (lowest stricture risk)**. **T pouch and extraserosal tunnel** provide antireflux when wanted.

Camey II —

- **Modification of the original tubular Camey: detubularised and folded** to abolish peristalsis.

65 cm ileum isolated, urethral point pre-identified → bowel restored, mesenteric trap closed → opened on antimesenteric border except at urethral point (incision deviates to mesenteric border there) → **transverse U**, medial borders sutured → fingertip opening, ileal plate to pelvis, **urethral anastomosis** → **Le Duc ureteric implantation** → plate folded and closed → U ends anchored to pelvic floor.

- **Barre Z modification:** shorter bowel, greater capacity, possibly better continence. **Stapled (GIA — gastrointestinal anastomosis) version** for speed.

Orthotopic Kock ileal reservoir (hemi-Kock) —

- **Origin:** continent cutaneous Kock with **two intussuscepted nipples** (afferent antireflux, efferent continence) → orthotopic form keeps only the **afferent nipple**. S-shaped variant (Schreiter and Noll).
- **Early results excellent** (>500 cases; ureteroileal stricture <3%), but **late afferent-nipple complications: stones on exposed staples, ostial stenosis (ischaemia), extussusception or prolapse** — often >10 years later and **silent until bilateral hydronephrosis or renal failure**. Now rarely used.
- **Figure 134.5 detail:** **61 cm terminal ileum = two 22-cm limbs** (U, opened adjacent to mesentery) + **proximal 17 cm** for a **5–7-cm afferent nipple**, made by stripping mesentery, intussuscepting with Allis clamps, fixing with **two staple rows** and staples to the back wall; dependent end left open for urethra.

Serosus-lined extramural tunnel (Abol-Enein and Ghoneim) —

- **Effective, durable antireflux:** >93% unidirectional, unobstructed flow; >1600 patients; **upper tracts preserved in ~90% at ≥10 years**.

40 cm ileum in a **W** → opened on antimesenteric border → medial flap edges sewn → serosa of each pair of lateral flaps joined with **3-0 silk seromuscular suture** → **two serous-lined troughs** → ureter laid in trough, **spatulated mucosa-to-mucosa anastomosis at the distal end** → mucosal edges closed over ureter → anterior wall closed side to side → dependent suture line reopened for urethral anastomosis.

- **Drawbacks:** needs **long ureters, cannot take dilated ureters**, possibly higher stricture risk.

Ileal neobladder (Hautmann pouch) —

- **Large-capacity spherical W (or M) reservoir** to maximise initial volume and reduce night incontinence.

~70 cm terminal ileum → urethral point marked with traction suture → M or W → opened antimesenteric except **5 cm at the traction suture (U-shaped flap)** → four limbs sutured → small full-thickness button excised, **urethral anastomosis tied from inside** → ureters implanted from inside at a convenient site → anterior wall closed.

- **Ureteric technique changed:** Le Duc antireflux initially → strictures common → **since 1997 freely refluxing open end-to-side** into short tubular segments at each end of the W → **ureteroileal stenosis fell from 9.5% to 1%**.
- **Hautmann vs Studer — the capacity trade-off:** larger initial capacity → **earlier continence**, but **more late retention and electrolyte reabsorption**. A **40-cm modification** mitigates this. Classic W could not take short ureters; **leaving one or both W ends long** solves it.

Studer pouch —

- **Long afferent isoperistaltic tubular ileal limb** — functionally prevents reflux during Valsalva voiding. **One of the most popular orthotopic diversions in the United States.**

- **Advantages:** simple, no staples, accommodates short ureters; reservoir uses the optimal **Kock double-folded U**.
- **Outcomes:** excellent continence and renal preservation, ureteroileal stricture <3%; most void without catheterisation up to 20 years; among ≥10-year survivors about a fifth had late complications, **pyelonephritis the commonest**.

Isolate terminal ileum 15–20 cm proximal to the ileocaecal valve → **distal mesenteric division in the avascular plane between ileocolic and terminal superior mesenteric branches**; proximal division kept **short** (broad blood supply) → **discard a small mesenteric window and ~5 cm of ileum proximally** (mobility) → stapled bowel anastomosis → **40–44 cm reservoir (two 20–22-cm limbs)** + proximal afferent limb (longer if ureters short) → close afferent end → open ~2 cm from mesentery → oversew in two layers of **running 3-0 polyglycolic acid** → fold in the opposite direction → close anterior wall → spatulated **end-to-side ureteroileal anastomoses (interrupted 4-0), stented** → urethral anastomosis with pre-placed sutures (dependent end of suture line or separate buttonhole).

STUDER AFFERENT LIMB LENGTH — TEXT VERSUS FIGURE

Source	Afferent limb	Total isolated	Reason for difference
Body text	15 cm (original description 20 cm)	54–56 cm stated	Studer's group later advocated a shorter afferent limb ; the figure shows it, and only it fits the 54–56 cm total
Figure 134.8	10–12 cm	44 + 10–12 = 54–56 cm	

T-pouch modification (Stein and Skinner) —

- **Purpose:** antireflux **without the Kock nipple's late complications**, with flexibility for dilated or short ureters; a modification of the serous-lined tunnel.

Divide ileum between afferent segment and the 44-cm reservoir segment → **anchor the distal 3–4 cm of the 15-cm afferent limb** in a serosal trough between the two 22-cm limbs → open **mesenteric windows of Deaver** between arcades of the T limb → **3-0 silk** serosal sutures through the windows fix the limb (preserving its blood supply) → at the afferent ostium, divert the U incision laterally and cephalad → **wide ileal flaps closed over the limb in two layers (flap valve)** → interrupted mucosa-to-mucosa ostial anastomosis (3-0 polyglycolic acid) → remainder as Studer.

- **Tapering the distal afferent limb** (early versions) was abandoned — associated with **late ostial stenosis**.

Padua pouch (vescica ileale padovana, VIP) —

~**42 cm distal ileum**, ~20 cm from the ileocaecal valve → urethral point **12–14 cm from the distal end**, usually most dependent → **two layers of folding** of posterior plate in a **spiral**, funnelled bladder neck → ureters spatulated through the posterior wall, **Le Duc/Camey implantation** with interrupted 4-0 Monocryl, stented → anterior wall closed with two sero-serous running sutures.

- **Robotic totally intracorporeal Padua:** 30-day complications ~44%; 2-year continence ~73% day, ~56% night.

Colon and Ileocolic Pouches

- **Indication for all-colon neobladders:** multiple small-bowel resections or diseased ileum. Colon is **less distensible and generates high-pressure waves** → **incontinence**, so the **initial volume must be larger**; adding ileum mitigates this.

Orthotopic Mainz pouch (Mainz III; "mixed augmented ileum and zecum") —

Originally continent cutaneous → orthotopic. **10–15 cm caecum + 20–30 cm ileum** → ascending ileocolostomy → whole segment opened antimesenteric, **ileocaecal valve sacrificed**, appendicectomy → **W** (first limb caecum, others ileum), three adjacent limbs sutured as posterior plate → **tunnelled ureterocolonic**

(submucosal) implantation in cephalad caecum → buttonhole in dependent caecum for urethra → closed side to side (cystostomy tube and stents out).

Right colon pouch (orthotopic) —

Whole right colon and caecum isolated → transverse ileocolonic anastomosis → ileal stump closed at the valve → opened along anterior taenia, **sparing proximal 2–3 inches of caecum** → appendicectomy → **antireflux ureteric implantation** → **Heineke–Mikulicz** fold and closure.

Sigmoid pouch (Reddy) —

- **Redundant sigmoid is usually available**; concerns: **distal colonic vascularity** (internal iliac branches interrupted at cystectomy) and temporary **frequent stools or rectal urgency** after sigmoid colectomy.

35 cm descending and sigmoid colon on the **inferior mesenteric artery** in a U → medial taenia incised to just short of the urethral point → medial limbs sewn → **tunnelled antireflux ureteric implantation** → dependent button for urethral anastomosis → closed side to side.

- **Da Pozzo modification**: fully detubularised and folded **Heineke–Mikulicz** for a more spherical reservoir.

ORTHOTOPIC NEOBLADDER DESIGNS COMPARED

Design	Segment	Shape	Ureteric implantation	Distinguishing advantage	Distinguishing drawback
Camey II	Ileum 65 cm	Transverse U (Barre: Z)	Le Duc — antireflux	Detubularised successor to Camey I	Le Duc strictures
Hemi-Kock	Ileum 61 cm	Double-folded U	Intussuscepted nipple — antireflux	Excellent early results	Late nipple stenosis, extussusception, staple stones
Serosus-lined tunnel	Ileum 40 cm	W	Extramural serosal trough — antireflux	Durable antireflux; tracts preserved long term	Needs long, non-dilated ureters
Hautmann	Ileum ~70 cm	W / M	Refluxing end-to-side (since 1997)	Largest initial capacity → earlier continence	More late retention and reabsorption
Studer	Ileum 40–44 cm + afferent limb	Double-folded U	Refluxing, isoperistaltic afferent limb	Simple, no staples, short ureters ; stricture <3%	Free reflux on cystogram (clinically benign)
T pouch	Ileum 44 cm + 15-cm T limb	Double-folded U	Tunnelled afferent limb (flap valve) — antireflux	Accepts dilated or short ureters	More secondary procedures than Studer (randomised trial)
Padua (VIP)	Ileum ~42 cm	Spiral, funnelled neck	Le Duc/Camey — antireflux	Anatomical funnel; robotic intracorporeal	—
Mainz III	Caecum 10–15 + ileum 20–30 cm	W	Submucosal tunnel	Ileocolic compliance	Loses ileocaecal valve
Right colon	Right colon + caecum	Heineke–Mikulicz	Antireflux	No ileum needed	Colonic pressure waves
Sigmoid (Reddy)	Descending/ sigmoid 35 cm	U (Da Pozzo: Heineke–Mikulicz)	Tunnelled antireflux	Redundant, available; spares ileum	Poor compliance ; distal vascularity; bowel urgency

Continent Cutaneous Diversions

- **Intussuscepted nipple valves are now historical**; the main approaches are the **Indiana pouch** and the **right colon pouch with (neo)appendicoumbilicostomy**.

CONTINENT STOMA SITE: UMBILICUS VERSUS LOWER QUADRANT

Umbilicus	Lower quadrant (through rectus, below "bikini line")
Lower stomal stenosis , especially appendiceal stomas; easiest for paraplegics (no transfer or undressing); near-invisible in a recessed umbilicus; patient should wear a medical alert bracelet	Concealable under clothing

EFFERENT CONTINENCE MECHANISMS FOR CONTINENT CUTANEOUS DIVERSION

Mechanism	Principle	Strengths	Limitations
Appendix tunnelled into caecal taenia (Mitrofanoff)	Flap-valve compression in submucosal tunnel	Simplest ; does not elongate or kink; excellent continence	Absent after appendectomy; may be too short ; admits only 14–16-Fr catheters (mucus clears better with 18-Fr); needs length $\geq 5-6$ cm and calibre ≥ 12 -Fr
Mitchell caecal lengthening	Tube of proximal caecum stapled in continuity with the appendix	Adds length; wider caecal stoma — less stenosis	Stapler must pass through mesoappendix window to protect blood supply
Yang-Monti ileal tube	2–3 cm ileum opened near the mesentery, retubularised transversely over a catheter; two joined for length	Substitute when appendix absent or unsuitable; well vascularised	Uses more bowel
Tapered ileum tunnelled into colon	Tapered ileal channel	Independent blood supply; no length limit	Further reduces absorptive surface
Tapered/imbricated terminal ileum + ileocaecal valve (Indiana)	Plicated valve + neourethral tapering	Reliable ; simple	Loss of ileocaecal valve → short-term bowel frequency (usually adapts or responds to drugs)

General procedural methodology —

Complete pouch → **fill with saline** → remove catheter from channel → **compress lightly to check for leaks and continence** → **re-catheterise to confirm easy passage** (inability to catheterise may mean reoperation) → **fix reservoir to the anterior abdominal wall** to prevent migration, false passage or kinking.

Indiana pouch —

- **Rowland, Indiana University, 1987: buttressed ileocaecal valve** able to withstand repeated catheterisation — improved on Gilchrist's intact-caecum reservoir.
- **Evolution**: imbrication initially along the whole ileal segment → found necessary **only at the valve** (continence zone confined there on pressure profiles) → partial colonic opening left enough caecal peristalsis to overcome the valve → **whole right colon marsupialised and folded Heineke-Mikulicz**. **Florida and Miami pouches** are minor variations.

~10 cm terminal ileum + entire right colon to the right/middle colic junction → appendectomy, cauterise the appendiceal fat pad at the ileocaecal junction → open whole right colon antimesenterically → **ureterotaenial implants** → **double imbrication of the valve: interrupted non-absorbable Lembert sutures, two rows ~8 mm apart over 3–4 cm, over a 16-Fr catheter** (Miami: purse-strings; Tampa: apposing Lembert

sutures) → taper remaining ileum with stapler → **imbricate while the pouch is still open** (see valve closing) → close Heineke–Mikulicz → stents via suprapubic tube or stab wound → rotate pouch, bring neourethra through umbilicus or rectus → **flush stoma to V-flap**.

- **Miami/Florida difference:** whole ascending colon + right third to half of transverse colon + 10–12 cm ileum, folded as an inverted U — **identical in effect to Heineke–Mikulicz**.
- **The Indiana pouch remains one of the most reliable catheterisable reservoirs.**

Right colon pouch (authors' preferred continent cutaneous diversion) —

- **Based on the Penn pouch** — the first continent diversion using the **Mitrofanoff (1980) principle** (ileocaecal patch, appendix and caecal button as the continence mechanism).
- **Advantages:** intact terminal ileum as afferent limb → **ureteroileal, not ureterocolic**, anastomosis (easier, theoretically fewer strictures); **intact ileocaecal valve = natural antireflux**; submucosal appendix = Mitrofanoff continence (**up to 100%** at medium term); appendix does not elongate or kink. Unsuitable appendix → **Yang–Monti neo-appendicoumbilicostomy**.

Mobilise right colon along the line of Toldt beyond the hepatic flexure → divide ileum **5–10 cm proximal to the valve** and colon **just proximal to the middle colic artery (~30 cm of colon for capacity)** → open appendix to take **≥12-Fr** → mesoappendiceal windows → **4–5 cm incision through anterior taenia to mucosa** → appendix laid in the trough, taenia closed over it (**3-0 silk through windows**) → detubularise colon, deviating away from appendix → fold inverted U, close back wall and pouch **Heineke–Mikulicz** → **bilateral end-to-side ureteroileal anastomoses over double-J stents tied to the pouchostomy tube** → fix caecum to posterior rectus fascia around the stoma → mature appendix at the umbilicus.

CONTINENT CUTANEOUS RESERVOIRS: INDIANA VERSUS RIGHT COLON POUCH

Feature	Indiana (Miami/Florida)	Right colon pouch (Penn-based)
Continence mechanism	Imbricated ileocaecal valve + tapered ileal neourethra	Appendix tunnelled in taenia (Mitrofanoff); Yang–Monti if absent
Ileocaecal valve	Used as the outlet (lost to bowel)	Preserved as the ureteric antireflux
Ureteric anastomosis	Ureterocolic (taenial) implants	Ureteroileal via intact terminal ileum
Reservoir	Entire right colon, Heineke–Mikulicz	~30 cm right colon, Heineke–Mikulicz
Key caveat	B12 loss (terminal ileum); bowel frequency	Narrow channel (small catheters); needs adequate appendix

Use of Minimally Invasive Techniques for Continent Diversion

- **Randomised trials confirm safety and feasibility of robotic cystectomy;** early intracorporeal diversion was difficult, so trials used **extracorporeal construction through a small midline incision** — which still allowed continent diversions. The **shift to totally intracorporeal diversion may worsen the drift to incontinent diversion**.
- **Intracorporeal neobladders (and small Indiana series):** outcomes comparable to open. Robotic intracorporeal Studer: **similar urodynamics** to open, yet **more pad use, wetness and catheterisation** despite shorter follow-up.
- **Still technically demanding:** longer operations, steep learning curve, same stay, **higher cost** → **centres of excellence;** long-term functional and quality-of-life data lacking (see Chapter 135).

FIGURE LEGENDS — SURGICAL TECHNIQUES (FIGURES 134.4–134.17)

- **Figures 134.4–134.11** illustrate the neobladders above (Camey II, Kock, serous-lined, Hautmann, Studer with T-pouch modification [Laramie Studio, 2012], Padua, Mainz, sigmoid). The **modified Camey II is**

folded three times (Z-shaped); in the Hautmann, ureters may be implanted **end to side or into the open ends** of the W.

- **Figure 134.12** Lengthened appendiceal stoma (Burns and Mitchell, 1990): GIA stapler passed through a mesoappendix window to protect blood supply; **caecal tube forms the stoma**.
- **Figure 134.13** Yang–Monti tube (Monti et al., 1997): segment opened a quarter of the way up one side → rectangular plate → closed over a catheter.
- **Figures 134.14–134.15** Indiana and Miami pouches (Benson et al., 1988; Olsson, 1989; Lockhart, 1987).
- **Figure 134.16** Penn-type pouch (Duckett and Snyder, 1987): **caecal button circumcised around the appendiceal origin**, mesoappendix preserved.
- **Figure 134.17** Right colon pouch (Warner and Chan, 2017): closed with running 3-0 polyglactin.

Concept — *Every named pouch is a variation on two independent choices: how the bowel is folded, which sets capacity and pressure, and how the ureters enter, which sets reflux against stricture. The modern favourites (Studer, Hautmann, right colon) converge on detubularised ileum with a direct or ileal-limb ureteric anastomosis, because the evidence rewards low stricture risk over reflux prevention.*

RESULTS AND COMPLICATIONS OF ORTHOTOPIC URINARY DIVERSION

- **Most post-cystectomy complications are diversion-related and diversion-specific**; they can be life-threatening.
- **Early = within 90 days; late = after. Lifelong follow-up** is essential because many complications appear years later.

Early Complications

- **Not diversion-specific**: bleeding, thrombosis, infection, cardiopulmonary events; **ileus** (from the bowel anastomosis common to all diversions); urine leak and ureteric complications.
- **Orthotopic vs conduit**: **no increase** in overall complications, stay or reoperation. **Continent cutaneous (usually colon)**: **more infection and fascial dehiscence**.
- **Urine leak**: commoner with continent diversion (long suture lines); usually **settles with catheter, stents and drains** unless a urinoma forms.
 - **Undrained leak → percutaneous drain and/or bilateral nephrostomies, not early open repair** (enterotomies, fistula). Fewer than ~5% need percutaneous intervention.
- **Stoma-specific (cutaneous only)**:
 - **Efferent limb necrosis** (rare, ischaemic) → **flexible endoscopy** to find the level; **ischaemia below the fascia → surgical exploration**.
 - **Stomal complications, chiefly stenosis**: up to ~20% early, but **more common after 90 days and after radiation**.

Late Complications

- **Confounders**: radiation, diabetes, comorbidity, tumour recurrence, adjuvant or salvage therapy.
- **Not diversion-related**: bowel obstruction, **ventral hernia**, thrombosis, cardiovascular disease. **Late complications accumulate over time → report as Kaplan–Meier estimates, not crude percentages**.
- **Ventral hernia is common** — partly from **straining to empty the neobladder**, plus age- and smoking-related fascial weakness; **dehiscence higher with cutaneous reservoirs** (infection, longer surgery).
- **Diversion-specific late complications**: incontinence, retention, infection, ureteroileal or afferent-limb obstruction, urethral stricture, upper-tract and pouch stones, vaginal fistula, pouch rupture — **most managed endoscopically; cutaneous stomal stenosis usually needs open revision**.

Urinary infection

- **Asymptomatic colonisation is very common**, especially with self-catheterisation; infection incidence similar for conduit and neobladder.
- **Symptomatic infection is uncommon after 90 days**; cumulative febrile infection low (Ulm), concentrated in the **first year** and in **poor emptiers** (several-fold higher than good emptiers).
- **Bowel mucosa, unlike urothelium, cannot inhibit bacterial growth.**

BACTERIURIA AFTER CONTINENT DIVERSION — TREAT OR NOT

Do not treat	Treat
Asymptomatic bacteriuria — treatment breeds resistance and is usually unnecessary	Symptomatic infection; virulent organisms (Proteus, Pseudomonas) even if asymptomatic — risk renal deterioration

- **Febrile infection or pyelonephritis after the first few months → look for obstruction, stones or incomplete emptying.**
- **Recurrent symptomatic infection with anatomy excluded → daily suppression with nitrofurantoin, a sulfonamide or methenamine.**

Pouch perforation

- **Rare** (low outlet resistance); risk higher with **prior radiation or chronic retention. Potentially fatal**: acute abdominal pain, distension, sepsis.
- **Cystogram is usually diagnostic**; stable patient → **catheter drainage ± percutaneous drains.**

Ureteroileal stricture

- **Same rate as ileal conduit**; determined by anastomotic type — **direct end-to-side (Leadbetter) or Wallace with interrupted fine absorbable sutures: lowest, ~3–6%.**
- **Usually within months, but can appear after 10 years → lifelong surveillance.**
- **Radiation is not an independent risk factor** on multivariable analysis (USC); **surgical technique is the main determinant.**
- **Often silent until bilateral hydronephrosis or renal failure.** Suspect on CT or ultrasound; confirm with **pouchogram (reflux), retrograde pyelography or antegrade nephrostogram.** CT is often misread by those unfamiliar with the pouch.
- **Obstructing antireflux segment → endoscopic incision** (the afferent limb can be hard to find). **Endourology succeeds far less often than open repair; retreatment exceeds 50%.**

Pouch stones

- **Very common in the Kock neobladder** (staples in the nipple), rising steadily with time; **rare in Studer, Hautmann, Indiana and appendicoumbilicostomy pouches** (all absorbable suture).
- **Metal GIA-stapled neobladders**: stones appear within 1–2 years in a few percent (stapled Camey II ~4.5% at 5 years) and will rise with follow-up; endoscopically treatable but costly and troublesome.

Pouch–vaginal fistula

- **Female-specific; difficult to repair**; incidence **5–10%** in larger series. **Risk ↑ with anterior vaginal wall excision and prior radiation.**
- **Prevention**: vagina-sparing when oncologically safe; **watertight vaginal cuff closure; omental flap between vagina and neobladder**, secured either side of the urethral anastomosis.
- **Detection**: suspected early leak → **lateral cystogram before catheter removal; persistent significant incontinence after a few months → exclude fistula** (pelvic examination ± methylene blue in the pouch).

- **Management:** beyond the first weeks, **will not heal with catheter or nephrostomy drainage alone** → transvaginal repair (variable success) → transabdominal repair or **conversion to cutaneous diversion**.

Continence

- **Results vary with** age, sex, prior radiation or prostate surgery, technique and experience — and with **definition, data collection method and follow-up**. **Validated patient-completed questionnaires are ideal** but imperfect: the **Bladder Cancer Index (BCI)** covers all diversions and captures leakage, but **not severity or pad use**. **Compare series with caution**.

Continent cutaneous stomal incontinence

- **Rare:** older series up to ~15%; contemporary USC — almost all continent by 90 days.
- **Persistent leakage = failed continence mechanism** → **surgical repair**. **Minor leakage** → **submucosal bulking agent** (e.g., Coaptite) at the pouch entrance. **Colon pouches** → **urodynamics** (high-pressure contractions).

Orthotopic continence — pattern and risk factors

- **Time course:** **daytime continence improves over the first 6–12 months** (most by 3–6 months); **night-time continence improves more slowly, into the second year**.
- **Pooled daytime incontinence ~13%.**
- **Risk factors for daytime incontinence:** **age >65, colonic segments, no nerve sparing** (some series), **hysterectomy, diabetes**.
- **Questionnaire-based male data:**
 - Mailed BCI survey (Studer or T pouch, ≥12 months): most used pads at least sometimes, but daytime pads were often small and dry or barely wet; **night-time pad use far higher**; older age and diabetes predicted worse outcome; **bother was generally low and correlated poorly with measured incontinence**.
 - Prospective pictorial pad questionnaire (USC): **daytime continence peaked ~92% at 12–18 months**; **night-time ~51% at 18–36 months**; age >65 and diabetes worse; only ~10% ever self-catheterised.

Nocturnal incontinence

- **One of the most bothersome sequelae; up to 50%.**
- **Mechanisms:**
 - **No neurological feedback or sphincter–detrusor reflex.**
 - **Lower sphincter tone and urethral sensitivity at night**; nerve sparing preserves **membranous urethral sensation**, so urine entering it triggers reflex or voluntary sphincter contraction — why nerve sparing helps day and night.
 - **Physiological nocturnal diuresis:** bowel mucosa **secretes water**, reversing nocturnal concentration; **lessens with time**.
- **Management:** **empty completely at bedtime, alarm to wake 2–3 times**; improvement continues to ~24 months; rarely a major quality-of-life problem. **Nightly desmopressin** may reduce nocturia (small study).

Women

- **Uterine preservation** → **better daytime continence**; better function also with uterine preservation plus nerve sparing.
- **Neurovascular tissue runs medial to the ureters, lateral to the cervix and paravaginally** → keep vaginal wall dissection **no more dorsal than 2 and 10 o'clock**. Benefits: longer urethra, **higher urethral closing pressure**, better continence, possibly better sexual function.

- **Validated questionnaire in women:** about a quarter leaked frequently by day and a third at night; **~60% self-catheterised at least daily** — likely closer to reality than chart reviews.

Managing incontinence after orthotopic diversion

- **Wait for the plateau — 12–18 months** — before evaluating.
- **Pelvic floor physiotherapy with biofeedback** helps initial continence. **Urodynamics** to confirm capacity without pressure waves, **especially colonic pouches**.

STRESS INCONTINENCE AFTER NEOBLADDER: MEN VERSUS WOMEN

Option	Men	Women
Bulking agent	For low closure pressure / low Valsalva leak pressure; marginal results	Minimally invasive; minimal symptoms only; success <50% , poorly durable; non-absorbable agents may do better
Definitive	Artificial urinary sphincter or male sling	Pubovaginal sling — more effective than bulking, but most will then need intermittent catheterisation
Safety rule	No blind passage of needles or sling material into the pelvis (pouch or adherent bowel at the symphysis) → infrapubic bone anchors or prepubic approach	

TABLE 134.1 RESULTS WITH ORTHOTOPIC DIVERSION IN SERIES WITH PREDOMINANTLY OR ONLY MALE PATIENTS

Rebuilt: series grouped by reservoir; ranges across series. Sources span Thüroff 1986 to Nayak 2018.

Reservoir group	Day continence	Night continence	Intermittent catheterisation (IC)	Antireflux mechanism
Ileal, detubularised — Studer, Camey II, Hautmann W, S pouch, unspecified ileal (chart review)	90–98%	74–95%	0.5–9%	Isoperistaltic limb, Le Duc, chimney or none
Serosal-lined extramural	93%	80%	2%	Serosal-lined tunnel
Kock ileal	87–98%	80–86%	8–32%	Nipple valve
T pouch	87%	72%	25%	T limb
Ileal, validated questionnaire (Studer or T pouch)	83–92%	47–51%	~10%	Partly T limb
Colon-containing — Mainz, right colon, Le Bag, sigmoid	89–95%	10–86% (sigmoid lowest)	0–37%	Submucosal tunnel, Le Duc

- **Across series:** mean age ~47–70 years; mortality 0–6%; diversion-related complications early ~4–15%, late ~5–29% (definitions vary).
- **What the table shows:** daytime continence is uniformly high (~83–98%); night-time is where designs and data methods diverge.
- **Pattern:** validated questionnaires roughly halve reported night continence compared with chart-review series — the method, not the pouch, drives the difference.
- **Pattern:** antireflux valves (Kock nipple, T limb) carry the highest self-catheterisation rates among ileal designs; sigmoid has the worst night continence, consistent with colonic compliance.
- **Trap:** in these tables "IC" means intermittent catheterisation, not ileal conduit.

TABLE 134.2 RESULTS OF ORTHOTOPIC DIVERSION IN SERIES LIMITED TO FEMALE PATIENTS

Rebuilt: series grouped by data-collection method. Sources span Stenzl 2001 to Bartsch 2014.

Series type	Day continence	Night continence	IC	Antireflux
Chart review / clinician-reported — ileal, Studer, serous-lined tunnel	57–92%	45–72%	12–35%	None or serous-lined
Validated patient questionnaire — Studer/T pouch, Hautmann	30–82%	35–76%	58–62% (39–45% wholly dependent on IC to void)	None or T pouch

- **Across series:** mean age ~54–66 years; mortality 0–2.5%; early complications 5–32%, late 5–35%.
- **What the table shows:** women achieve **comparable daytime continence** in clinician-reported series, but **far higher catheterisation dependence than men** (roughly 12–62% vs 0–37%).
- **Pattern:** validated questionnaires expose **much lower continence and majority self-catheterisation** — hypercontinence (retention) is the female problem.
- **Trap:** assuming women's main functional failure is leakage; it is **failure to empty**.

Urinary Retention

- **Presentation:** occasionally acute; more often **recurrent infection, new overflow incontinence**, palpable pouch or **new hydronephrosis**.
- **Late, rising event:** in men (USC), freedom from retention **93% at 3 years, 88% at 5 years, 79% at 10 years** — reported rates depend heavily on follow-up length.
- **Risk factors (men): excessive bowel length and prostate-sparing technique.** Anastomotic position (end of suture line vs separate buttonhole) — suggested cause of folding, but **a separate opening conferred no protection**.
- **True neobladder-urethral anastomotic stricture is rare** (~1% in Ulm without recurrence; a few more distal strictures).
- **Work-up:** rectal or vaginal examination and cystoscopy — exclude anastomotic stricture or **tumour recurrence**.
- **Treatment = intermittent self-catheterisation; drugs do not help.** An obstructing **mucosal flap** may be incised endoscopically (rare but correctable).
- **Abdominal wall / incisional hernia:** Kaplan–Meier ~4% at 5 years, ~6% at 10 years (Ulm); up to ~13% at 3 years prospectively — **reduces ability to raise abdominal pressure to empty**.

Retention in women

- **Cause uncertain — mechanical, not neurogenic:** video-urodynamics show the **pouch falling back into the wide pelvis → acute angulation of the posterior pouch-urethral junction**, often with **herniation through the prolapsed vaginal stump**; pelvic floor electromyography silent during voiding.
- After hysterectomy, emptying failure was common, mostly from **neocystocele**; dynamic MRI showed neobladder descent and a widened neocystourethral angle on straining.
- **Proposed preventive measures:** **omentum behind the reservoir**; rectal peritoneum sutured to the vaginal stump; **vaginal stump suspended from round ligaments**; **pouch dome suspended to the posterior rectus**.
- **Prophylactic sacrocolpopexy** (8 × 3 cm polypropylene mesh; small randomised trial) — markedly lower residuals and catheterisation. Authors' routine sacrocolpopexy **has not eliminated retention**.
- **Uterine and ligament preservation may substantially reduce retention**; lateral vaginal neurovascular preservation may aid sphincter relaxation. **No randomised trials** for these theories.

KEY POINTS: CONTINENCE AND URINARY RETENTION

- **Daytime continence develops over 3–6 months; ultimately 80–90% in both sexes.**
- **Persistent nocturnal incontinence 20–50%; improves beyond 12 months and with timed voiding.**
- Persistent cutaneous incontinence → **surgical revision of the efferent mechanism.**
- Continence influenced by **age, bowel segment, nerve sparing, prostate-sparing technique.**
- **Retention: 4–10% of men, 20–30% of women** (rises with follow-up — ~21% of men by 10 years in the USC series).

Concept — Outcomes after orthotopic diversion show a **sex asymmetry**. **Men fail by leaking, especially at night, because the retained sphincter is weaker without its reflexes.** **Women fail by not emptying, because the pouch loses its posterior support and kinks at the urethra.** Continence treatment follows from that: **support the female pouch, and augment the male sphincter.**

FOLLOW-UP FOR PATIENTS WITH CONTINENT DIVERSION

- **No consensus regimen; imaging by intravenous urography (rare now), CT urography, MRI or ultrasound — United States favours cross-sectional imaging, Europe ultrasound.**

PHASES OF FOLLOW-UP

Phase	Target	Tool
Early — first 4 months	Ureteroileal strictures (technical error, ischaemic distal ureter)	Upper-tract imaging
Intermediate — 4 months to 3 years	Cancer recurrence ; also stones and obstruction	CT / cross-sectional , frequency risk-adapted to pathology
Long term — beyond 3 years	Pouch stones, late upper-tract obstruction, urothelial cancer of urethra or upper tracts	Imaging, cytology, examination

BOX 134.1 SUGGESTED FOLLOW-UP PROTOCOL FOR BLADDER CANCER PATIENTS AFTER CYSTECTOMY AND ORTHOTOPIC DIVERSION

Schedule	Content
Every 4 months in year 1, 6-monthly to 3 years, then annually	Physical examination including pelvic examination ; comprehensive metabolic panel; full blood count
Annual only	Urethral wash cytology (if CIS on pathology); vitamin B12; prostate-specific antigen (if incidental prostate cancer)
Imaging — pT3 or N+ and above	CT abdomen and pelvis with contrast + CT chest or chest radiograph every 3–4 months in year 1 , then 6-monthly to 3 years, then annually to 5 years
Imaging — pT2 or below	CT abdomen and pelvis + chest radiograph at 4 months , then 6-monthly to 3 years, then annually to 5 years
After 5 years — indefinitely	Annual upper-tract ultrasound , metabolic panel, full blood count

- **Box logic:** intensity scales with **stage** (recurrence risk), while the lifelong elements — **B12, metabolic panel, upper tracts** — track the **bowel and the ureteric anastomoses**, which never stop being a risk.
- **Cytology:** voided and, selectively, **urethral wash in men — especially with prostatic urethral CIS**; needs an experienced cytologist.
- **Women: bimanual examination and voided cytology** — urethral recurrence, though rare, is **invisible on CT**.

- **Cystoscopy for new retention, changed voiding pattern or haematuria.**

KEY POINTS: COMPLICATIONS FROM CONTINENT DIVERSIONS AND RECOMMENDED FOLLOW-UP

- **Early morbidity and mortality not increased** vs ileal conduit.
- **Undrained leak → percutaneous drains ± bilateral nephrostomies**; avoid early open revision.
- **Most late urological complications are endoscopically manageable.**
- **Regular imaging** for silent upper-tract obstruction and stones.
- **Physical and pelvic examination plus urethral cytology** for urethral recurrence.

Concept — *Follow-up shifts target over time: from surgical error (strictures), to cancer, to the slow consequences of putting bowel in the urinary tract (stones, B12, obstruction, urethral tumour). Because the late threats are clinically silent, surveillance never stops.*

Quality of Life After Continent Urinary Diversion

- **Health-related quality of life (HRQoL)** increasingly shapes diversion choice; **urinary incontinence and leakage are the most studied domains.**
- **Early comparative work (1980s)**: patients were generally **satisfied and adapted well**; also showed that **how data are collected changes results.**
- **Distress**: neobladder patients similar to matched controls at 1 year, yet nearly half of cystectomy patients report general distress. **Body image** is a recurring theme.
- **Multi-questionnaire comparison** (conduit, cutaneous Kock, neobladder): most reported good quality of life and little distress; **after age adjustment, no difference between diversions.**
- **Neobladder patients often report better body image** than conduit patients — partly cultural.
- **Methodological weakness dominates**: a systematic review found no randomised and almost no prospective studies, with **unvalidated or generic instruments, no baseline, cross-sectional and retrospective designs** → **no diversion shown superior on HRQoL**; other analyses agree.

HRQOL INSTRUMENTS

Type	Instruments	Limitation
Generic	Medical Outcomes Study Short Forms (SF-36, SF-12); EuroQol five-dimension (EQ-5D)	Insensitive to diversion-specific issues
Condition-specific	Functional Assessment of Cancer Therapy – Bladder (FACT-BL); European Organisation for Research and Treatment of Cancer bladder module (EORTC QLQ-BLM30); BCI	Weight symptoms unique to one diversion — biased for continent vs conduit comparison
Neobladder-specific	IONB-PRO (23 items)	Not applicable across diversion types

- **Recent evidence**: still cross-sectional and retrospective, but a systematic review **favoured ileal neobladder**, and a meta-analysis of EORTC QLQ-C30 studies found **better quality of life with neobladder.**
- **Large prospective cohort (Memorial Sloan Kettering Cancer Center; 14 patient-reported measures over 2 years)**: physical function dipped then **recovered by 6 months**; **social function rose with continent diversion**; **body image worse with incontinent diversion**; **global quality of life exceeded baseline in all by 6 months.**
- **Needed**: prospective, longer, validated disease-specific studies; a single cross-diversion instrument is very hard to build.

TABLE 134.3 SUMMARY OF COMPARATIVE URINARY DIVERSION HEALTH-RELATED QUALITY-OF-LIFE STUDIES SINCE 2000

Rebuilt: studies grouped by verdict (2000–2022; United States, Europe, Asia). In this table IC = ileal conduit, NB = ileal neobladder.

Verdict	Instruments used	Findings
No difference between IC and NB — the largest group (~10 studies)	SF-36, EORTC QLQ-C30, FACT-VCI, BCI, informal questionnaires	Including long-term BCI data and female-only series; some found both groups below population norms
NB better (~8 studies)	Mainly EORTC QLQ-C30 ± BLM30 , SF-36	Higher global HRQoL or physical functioning; several only marginal, slight or non-significant
Body image worse with IC	FACT-BL; multi-instrument prospective cohort	Lower body image with conduit; both recover most domains by 3 months except sexual function (both) and body image (IC)
NB worse in a domain	BCI	Worse urinary HRQoL with NB ; conduit and Indiana pouch — better urinary, worse sexual function than NB
Below population norm	SF-36	Lower mental scores in conduit patients

- **What the table is for:** showing that **verdict depends on instrument** — generic tools mostly find no difference, EORTC tools tend to favour neobladder, and the urinary-focused **BCI penalises the neobladder** for leakage.
- **Consistent signal:** **body image favours the neobladder; urinary function favours the conduit.**
- **Trap:** in Table 134.3, "IC" means **ileal conduit**, although the table footnote repeats the Tables 134.1–134.2 definition (intermittent catheterisation).

Summary

- **No clear overall quality-of-life advantage for continent diversion;** most patients **adapt to whatever diversion they have.**
- **Fully informed patients have less decision regret.**
- Given the choice, **most choose orthotopic** (natural, no stoma), and the option may encourage patients to **accept definitive treatment** — but expectations about **incontinence and self-catheterisation must be realistic.**

KEY POINTS: QUALITY OF LIFE AFTER URINARY DIVERSION

- Most comparative quality-of-life studies are **methodologically flawed.**
- **Evidence insufficient to call any diversion superior on HRQoL**, though most report **better body image with neobladders.**

Concept — *Quality-of-life comparisons fail because each diversion trades one burden for another: the conduit gives up body image, and the neobladder gives up urinary control. An instrument's verdict therefore reflects which burden it weighs, which is why informed choice, not a superior diversion, is the evidence-based endpoint.*